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Concept and demonstration of a fully coupled and dynamic exposure-response methodology for crowd evacuation numerical modelling in airborne-toxic environments

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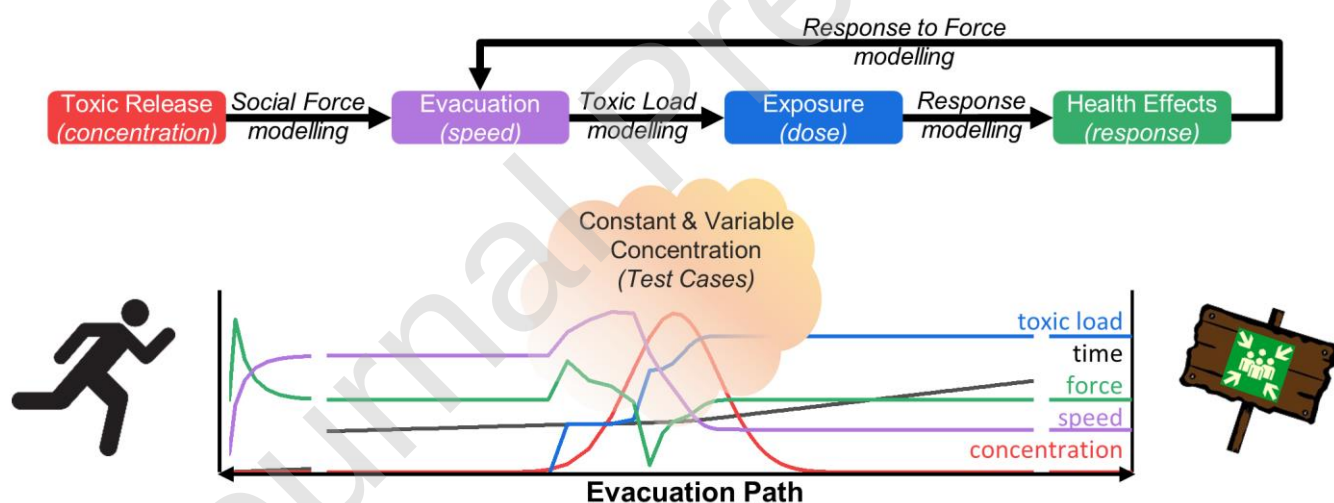
Concept and demonstration of a fully coupled and dynamic exposure-response methodology for crowd evacuation numerical modelling in airborne-toxic environments

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Graphical abstract



Highlights:

- A fully coupled and easily adaptable exposure-response-evacuation algorithm
- Theoretical demonstration of the algorithm on realistic H₂S case studies
- Profound differences with state-of-the art evacuation modelling approach
- Absence of the proposed coupling clearly leads to underestimated impact/risk

- New toxicological data are necessary to improve emergency response/preparedness

Abstract (~200 words):

Evacuation simulation plays an indispensable role when planning for emergencies and evaluating the consequences of disasters related to airborne toxics. Various attempts have been made to improve simulation of evacuation in toxic environments and to account for the varying concentration levels and the dynamic exposure. However, most studies neglect the reverse effect, how the exposure and dosage levels affect the physical and psychological state of an evacuee and consequently the evacuation path and process. In this work, a fully coupled exposure-response-evacuation and agent based algorithm is proposed, focusing on the H₂S airborne toxic. Accordingly, the dynamically estimated exposure of the evacuee affects, non-linearly, the evacuation speed and thus the overall evacuation. This coupling is challenging and thus it depends on the available epidemiological and toxicological data. Nevertheless, the diversity and advantages of the algorithm is successfully demonstrated over three case studies including single- and multi-agent in straight-path and building evacuation scenarios. For example, the building evacuation time increased by more than 50% by inclusion of the aforementioned coupling. In conclusion, herein, a gap on the evacuation modelling is addressed by a fully coupled methodology that could be easily adapted by safety engineers and further improved by researchers as more data become available.

Keywords:

Evacuation Modelling; Social-Force; Hydrogen Sulphide; Emergency Response; Risk

1 Introduction

Airborne toxic substances can potentially lead to adverse effects on the environment and human health, depending on their atmospheric transport, exposure to them, and their toxicity. In many cases, e.g. large accidental releases, the only viable safeguard against the hazard is the evacuation of the people to a safer place. Therefore, numerous evacuation models have been proposed to simulate such scenarios and cases for emergency planning and response (Kuligowski, 2016). These models provide insights into the evacuation process and assist on the analysis and assessment of the factors associated with crowd evacuations. Depending on the employed numerical approach, evacuation models can be classified into (Vermuyten et al., 2016): continuum, network-based, cellular automata, social force, and agent-based; ordered in increasing complexity and computational cost.

Continuum or fluid dynamics models simulate crowds as a whole (Henderson, 1971; Mnasri and Farhat, 2016), which limits their applicability to accurately simulating only high density crowds (Hughes, 2003). Network-based, cellular automata, and lattice gas models use discrete temporal and spatial scales to simulate the movement of individual evacuees according to a set of 'local rules'. While they operate in a similar manner, their difference arises in their physical interpretation of the evacuation processes. Cellular automata track agent movements through changes in "cells", while lattice gas models represent agents as actual particles. They are computationally inexpensive and have been used to study various phenomenon associated with emergency evacuations like the effect of obstacles (Varas et al., 2007), the friction effects, the clogging (Kirchner et al., 2003b) and the competitive egress behavior (Kirchner et al., 2003a). Social force models offer a continuous representation of the effects that rule the motion of the evacuees – the desire to reach a destination, the desire to keep a certain distance from obstacles and walls, and a certain distance between oneself and other evacuees – through an empirical force equation. Social force models have shown good agreement with experiments e.g. conducted on ants (Altshuler et al., 2005), but the resultant human behavior is homogenous, contrary to what is seen in real life as per the detailed reviews of (Zheng et al.,

2009) and (Vermuyten et al., 2016). Likewise, game theoretic models and approaches based on experiments with animals demonstrate the same challenge, although, they address the dynamics of escape panic efficiently (Vermuyten et al., 2016). Instead, agent-based models are able to incorporate inhomogeneous behaviors and interactions by modelling each evacuee individually, through a set of social constructs. This sustains them capable of simulating more complex evacuation processes, when combined with cellular automata (Braun et al., 2005) and social force models (Pan et al., 2007). Currently, agent-based models incorporating social force models are default option in many popular evacuation tools like the FDS+EVAC (Korhonen and Hostikka, 2010) and Pathfinder (Zeng et al., 2018).

Evacuation models require “awareness” of both the environment and the evacuees, in addition to the hazards or consequences in place. The inherent variability of the input information and its impact on the estimated health effects and evacuation outcome is well-known, thus many studies incorporated stochastic approaches for simulating evacuation (Georgiadou et al., 2007). Cai et al. (2010) considered the uncertainty of source location and proposed a framework to minimize the relative exposure level of building occupants. Georgiadou et al. (2010) identified the impact of the interactions introduced by the various protective actions and proposed an evolutionary optimization algorithm as an emergency response decision-making tool. Similarly, Gai et al. (2018b) accounted for the variability on exposure and dose by combining a human vulnerability model and accident probability to balance the relationship between cost and benefit in risk analysis. Sanchez et al. (2018) combined a local social vulnerability index, to classify the population, with pre-processed meteorological databases to define the most probable and conservative risk scenario.

Especially for airborne toxics, the evacuees’ experiences and exposure are dynamic, for example both the psychological and physical state of humans exposed to hydrogen sulfide (H_2S) vary and progressively degrade as the dosage increases (Xu and Fan, 2014). Therefore, the consequences associated to the different poisoning scenarios need to be considered for the individual risk assessment (Wu et al., 2019). Various attempts have been made to improve simulation of evacuation in toxic environments and to account for

this variability, however, most studies so far neglect the reverse effect i.e. how the exposure and dosage levels affect the physical and psychological state of an evacuee and consequently the evacuation path and process. For example, Lovreglio et al. (2016) employed an agent based model to estimate how the exposure levels of evacuees change based on the path taken. Liang et al. (2016) used also an agent based evacuation model and incorporated the fatality effect on the efficiency estimation of the evacuation. The evacuation efficiency and health consequences were calculated by Gai et al. (2018a) using a dynamic exposure approach, as well. One step further, Wan et al. (2014) applied the effects of toxic exposure on the ability of agents to evacuate based on the distance from the source and concentration levels (not dosage). Durak et al. (2015) simulated evacuation on vehicles, calculated the dynamic exposure of the drivers, and proposed a binary effect (true or false) on their driving capacities. A number of other studies proposed theoretical emergency response frameworks that integrate online information, via portable devices or networks of sensors, as a possible resolution (Xu and Fan, 2014; Scherer et al., 2018; Shuaib, 2018). To the best of our knowledge, there is one fire evacuation model that has proposed a fully coupled solution. Bae and Ryou (2015) developed a smoke effect model for representing the psychological pressure experienced when the evacuees are confronted with smoke or they move through the smoke screen.

To address this modelling gap, here a theoretical and fully coupled exposure-response-evacuation algorithm is proposed that improves the evacuation modelling in the case of airborne toxics. Current state-of-the art models omit the effect of exposure to a toxic agent on the evacuation process itself. Such simplifications have multiple impacts and cascading effects on evacuation modelling and emergency planning. For example, most agent models assume a maximum walking speed for the evacuees (Helbing et al., 2000). However, it might be significantly lower for agents exhibiting symptoms like eye or lung irritation (Stevens et al., 1999) or higher for agents exhibiting intense anxiety (Fiedler et al., 2008). A reduced walking speed could lead to longer evacuation times, higher dosages, more severe symptoms, further reduction of the walking speed, and ultimately misleading consequence analysis. The coupling of evacuation and exposure is challenging since its results depend on the available epidemiological and toxicological data. Therefore, aim of this work

is to present a methodology and its implementation algorithm rather than a universal solution. For the development and demonstration, we selected a specific toxic agent, H_2S , owing to its high toxicity (nearly instant death at $\sim 1,000\text{ppm}$), importance in oil and gas industry (there are fields with up to 42% content), and the available literature with toxicological information. The working assumption of this work, is that the proposed algorithm should have universal applicability in the condition that the required toxicological information is available.

2 Methodology

The proposed exposure-response-evacuation algorithm combines three models, fully coupled, and in a closed loop configuration (see Figure 1). The first model (section 2.1) deals with the exposure estimation of the agent depending on the concentration of the airborne toxic, which can be calculated by another dedicated model (e.g. Argyropoulos et al., 2017). The exposure model is based on the Boris and Patnaik (2014) approach that utilizes the ten Berge (1986) Toxic Load for time varying concentrations; although it requires keeping track of the exposed concentration levels its calculation is straightforward and fast (Argyropoulos et al., 2018). The second model (section 2.2) deals with the exposure-response relationship, herein specifically for H_2S , and classifies Toxic Load values into four ranges each with distinct medical symptoms, based on data from multiple toxicological studies. The third model (section 2.3) is inspired by the work of Bae and Ryou (2015) on the smoke effect on evacuating agents. Accordingly, it expands the Helbing et al. (2000) social force model with an additional term, named Toxic Force, to account for both the psychological and physiological effects of the airborne toxic. Furthermore, this model has been selected as basis because of it has been validated with actual data

To properly assess the concept of the proposed algorithm, it has been implemented in a simple agent-based simulation tool and it is demonstrated for multiple evacuation scenarios at a literature extracted building evacuation case (Argyropoulos et al., 2017).

2.1 Exposure Model

The incidence or severity of a toxic is a function of the exposure concentration and duration, it is typically referred to as dosage (TL_H), a concept put forward by Haber (as discussed in Witschi, 1999), and described by a simple equation which unfortunately does not describe exposure to all substances with satisfactory agreement:

$$TL_H = Ct \quad (1)$$

To overcome this, ten Berge transformed the linear expression to a power-law by introducing the toxic load exponent, n , a constant derived from experimental studies conducted on animals, to account for the variability of substances' toxicity (ten Berge, 1986). For time varying concentration levels, the integrated Toxic Load (TL_B) by:

$$TL_B = \int C(t)^n dt \quad (2)$$

In practice, the impact of the airborne toxic is the main concern, rather than the exact values of the toxic load, which is represented usually with limit and guideline Toxic Load or concentration values e.g. AEGL (Acute Exposure Guideline Levels), LD (Lethal Dose), IDLH (Immediately Dangerous to Life and Health). Therefore, Boris and Patnaik (2014) proposed a numerical approach to estimate whether one of the three AEGLs has been exceeded or not. In brief, using a time varying concentration profile, a loading rate is estimated at given time steps using the (ten Berge, 1986) power law equation (2). The cumulative loading rate is then normalized using each AEGL and forms the three new Toxic Loads, each varying from 0 to 1 i.e. being False or True respectively. When a Toxic Load is True (exceeds 1), this time step is considered the onset of the respective AEGL.

Here, we adapt the above numerical approach for the exposure model but we modified it in order to estimate one continuous Toxic Load (TL) varying from 0 up to the number of considered levels (e.g. 3 for the AEGL). The modification is based on a simple algebraic transformation, thus not described further, and it

respects both the continuity of the exposure profile and the original approach. Furthermore, instead of the AEGLs we employ data directly for toxicological studies, as proposed by (Hawkins et al., 2018), more details offered in the following section.

Noteworthy, the Boris and Patnaik (2014) approach does not address the effect of human body detoxification, which is more apparent when the concentration is significantly reducing or zeroes in between exposures (Ride, 1995). Also, in the literature, there are more exposure models that could be used and one should select the most appropriate for the toxic in question (Yee, 2013).

2.2 Response Model for Hydrogen Sulfide

Hydrogen sulfide is a colorless gas with a characteristic odor of rotten eggs. It naturally occurs in the gases from sulfur springs, volcanoes, and in crude petroleum and natural gas. It is frequently encountered in natural gas production industry and may be released to the environment as a result of its operations. Hydrogen sulfide toxicity targets mainly the respiratory tract and nervous system even at fairly low concentrations (OSHA, 2019), thus, tends to be a problem for the related industry and nearby communities.

At the same time, the global demand for abundant and cheap energy pushes the exploration and extraction of even more “sour” reservoirs (industry term for natural gas with more than 4 ppmv of H_2S). For these reasons, its toxicological profile is studied for decades and updated often (EPA, 2003; NRC, 2003; ATSDR, 2016).

The exposure and metabolic pathways for H_2S are fairly understood (ATSDR, 2016) but there is no standard exposure-response model, obviously, owing to the high complexity of the employed phenomena. Consequently, the abundant toxicological studies and risk analysis frameworks can be employed to develop a deterministic response model based, however, on qualitative inputs. Similar studies (e.g. Boris and Patnaik, 2014) have employed the AEGL, which are a set of time and concentration values (i.e. exposures) that describe human health effects from rare exposure of general population (including susceptible individuals)

to airborne chemicals (NRC, 2003). They consist of three distinct health effect levels (AEGL 1 to AEGL 3; i.e. response) and are provided for five time intervals for each level. However, many studies have criticized their application beyond the general population. For example, Hawkins et al. (2018) note that these guidelines are protective in nature and their use for predictive modeling can exaggerate consequences by orders of magnitude higher than those when toxicological data is used. Guidotti (2010) claims that empirical evidence strongly favors higher exponents n (see equation 2) for the lethality assessment than many of the current risk models, implying the AEGL. Additionally, many review studies (Guidotti, 2010, D. S. Gillies, 1999 #3825) and reports (EPA, 2003; ATSDR, 2016) confer that the exposure relationship (concentration vs time) for acute effects, particularly the ones related to the central nervous system and respiratory system, can be very steep. Indeed, this relationship is obvious in Figure 2 along with the significant differences between them e.g. “Death” curve by (Guidotti, 1996) and the AEGL.

On the other hand, herein the emphasis is on the evacuation modelling methodology rather than the response relationships and the toxic gas per se. To put it differently, the selected response relationships need to describe specific physiological and/or psychological responses, as well, that should be introduced into an evacuation model. With this in mind, the qualitative dosages (Table 1) described by (Guidotti, 2010), adopted by (OSHA, 2019) and endorsed by (ATSDR, 2016) qualify with the above criteria (i.e. odor, irritation of eyes and lungs, and pulmonary edema), and the physiological and psychological responses employed are discussed in the next section. These dosages, however, are missing a full concentration-time relationship and have only indicative values. Here, the full relationships have been constructed by assuming that they will share a similar steep profile to the AEGL values. Figure 2 illustrates the fitted power law curves on the selected dosages and the subsequent ranges of the modified Toxic Load (TL), which signals the onset of each response. Hereafter, the three TL ranges (i.e. 0-1, 1-2, and 2-3) are referred as bands and the responses as symptoms.

2.3 Modified Social Force – Evacuation Model

Using the concept proposed by Levin (1951), that behavioral changes are guided by social fields or social forces as a motivation, Helbing proposed a social force model for pedestrian dynamics (Helbing, 1994). According to that model, every individual, called agent, is subjected to three forces: The motivational force to represent the inherent nature of an agent to reach a destination, the repulsive force to enforce territorial effects on the agent, and the attractive force to account for the formation of pedestrians' groups. Later, the social force model was improved (Helbing et al., 2000) to simulate the dynamic features of human behavior induced during panic situations, like people moving faster than normal, pushing and physical interactions between people, uncoordinated passing of bottlenecks, jams, arching and clogging at exits, and tendency of mass behavior. Hence, the motion of each agent is described by the bellow differential equation:

$$m_i \frac{dv_i}{dt} = m_i \frac{v_i^o e_i^o(t) - v_i e_i(t)}{\tau_i} + \sum_{j(\neq i)} f_{ij} + \sum_w f_{iw} \quad (3)$$

where, subscripts i, j denote the agents, and w the walls, superscript o represents a fixed (desired) value, m [kg] is the mass, v [m.s⁻¹] is the velocity, e is the direction vector, τ [s] is characteristic time constant (here assumed equal to 1 s), and f [N] is a force. More specifically, on the right hand side, the first term is the motivational force of an agent to achieve the desired velocity and direction (typically set to 1.35 m.s⁻¹ and towards the exit). The second term accommodates the agent-agent interactions and the third term represents the interaction of the agent with walls.

To introduce the effect of the toxic gas on the motion of an agent, a new force term, f_{tf} (Toxic Force), is proposed following the example of (Bae and Ryou, 2015) on the smoke effect model. The Toxic Force, should accommodate the physiological and psychological "stresses" on an agent due to the cumulative exposure. This means that the agent will move with a different velocity than the desired one, v_i^o , in the motivational

force term. Instead of adopting a direct alteration of the desired velocity, the addition of this new force, analogous to the motivational force, was considered consistent with the original social force model, and more transparent in its implementation:

$$f_{tf} = m_i \frac{v_i^o(-e_i^o) - v_i^{tf}(-e_i^0)}{\tau_i} \quad (4)$$

where v_i^{tf} is the average velocity of an agent's exposure to the toxic gas, defined as a function of the cumulative toxic load (TL) for each of the TL bands (Table 1), and discussed hereafter.

The first band ($0 < TL < 1$) relays to the onset of the smell symptom. Following the extensive works of Bhambhani et al (Bhambhani and Singh, 1991; Bhambhani et al., 1994, 1996a; Bhambhani et al., 1996b, 1997) and Fiedler et al. (2008) who subjected humans to an H_2S concentration of up to 15 ppm, while they were exercising, there is no detrimental effect on the ability to move. At the same time, Fiedler et al. (2008) reported increased anxiety levels for concentrations at around 5 ppm, compared to 0.05 ppm and 0.5 ppm. On this basis, we assume that the combination of the anxiety (psychological stress) with normal physiology will lead to an increased desired velocity i.e. 2 m.s^{-1} , that is higher than the average walking speed (of 1.35 m.s^{-1}) but still lower than the jogging speed (less than 2.68 m.s^{-1}) (Helbing et al., 2000).

The second band ($1 < TL < 2$) deals with onset of the irritation symptoms, that entails coughing, throat irritation, lung irritation and eye irritation. To the best of authors knowledge there is no related study, however, (Stevens et al., 1999) studied the walking speeds of patients with Chronic Obstructive Pulmonary Disorders (COPD), who suffer from similar to above irritations. That study reported a maximum achievable walking speed of around 1 m.s^{-1} and this is assumed to be the maximum attainable speed on an agent at the upper limit of the band.

The last band ($2 < TL < 3$) incorporates the onset of the pulmonary edema along with the overstimulation of the nervous system, leading to minimization of the walking speed and loss of consciousness respectively.

Given these points, two non-linear functions were fitted to the above data (using the Levenberg Marquardt method in OriginLab software) in order to obtain a continuous transition among the different walking speeds

as a function of the also continuous TL (see section 2.1), rather than employing an abrupt square function behavior (Figure 3).

3 Numerical Simulation

Coupling of these three models is essential to the proposed exposure-response-evacuation algorithm (Figure 1). The evacuation model (section 2.3) is the backbone of the algorithm and relies on the other two models to convey information obtained from a toxic gas release scenario. The exposure model (section 2.1) first reads the concentration of toxic gas at an agent's location and calculates the rate at which the Toxic Load accumulates at the given concentration. These rates are accumulated differentially into the Toxic Load using equation (2). Each agent's accumulated Toxic Load is then translated into evacuation velocities (Figure 3) through exposure-response data (section 2.2). The evacuation model (section 2.3) then uses the modified evacuation velocity to translate toxic gas release concentrations and exposure times into their effects on crowd evacuation.

In order to gain a better understanding of the proposed concept and demonstrate its algorithm, the latter was implemented in MATLAB (v2017a) and two stages. The first stage includes the exposure model, response model, and calculation of v_{tf} (TL). The second stage introduced these into an existing, 2D, and simplified evacuation tool entitled "Panic Simulator" that solves Helbing's social force model using an explicit Runge-Kutta (2,3) pair of Bogacki and Shampine with the built-in ODE23 MATLAB function (Schmidt and Spah, 2015). The algorithm was tested in three different test cases. The first case (A) aims to verify the calculation procedure for a single agent (80 kg mass) that is moving along a straight path of 300 m length while exposed at constant concentrations of H_2S . The second case (B) examines the evacuation of the previous agent while exposed at varying concentrations of H_2S in order to assess qualitatively the sensitivity of the proposed algorithm. Figure 4 presents graphically the scenario while the concentration levels and profiles for each of the A and B test cases are illustrated in Figure 5.

Finally, in the third case (C), the algorithm is tested on a realistic evacuation of 51 agents from a building for different and constant concentration levels. The full building layout is presented in Figure 6a, it is a real non-process building at a natural gas processing facility in Qatar, and it has been used in the past for similar studies from the authors (Argyropoulos et al., 2017). For clarity and due to the limited layout design capacity of the employed software (Panic Simulator), only one third of the building (see Figure 6b) has been included in this study while many of the architectural features have been simplified. This is not expected to influence any of the calculation steps nor results. The same figure presents the evacuation zones that have been identified and are presented and discussed in the next section. Finally, the inherent variability of the agents' (people) characteristics has been addressed by executing multiple simulations per concentration level while varying the body weight (using the radius), desired velocity, and exact position of each agent using Monte-Carlo sampling from a normal distribution of values as proposed in Korhonen and Hostikka (2010).

4 Results and Discussion

4.1 Test Cases A – Verification with single agent and constant concentration

In test case A, a single agent evacuates by walking along a straight path of 300 m length, while exposed to constant H_2S concentration levels of 0, 5, 140, and 450 ppmv (Figure 4). Initially, the agent is static and Figure 7 presents the temporal evolution of the agent's location (x), toxic load (TL), total social force applied (Σf), and velocity (v) for all four cases. In the absence of H_2S (Case A1), thus zero toxic load, but with the motivation to evacuate, a modest force is exerted on the agent to reach the desired velocity of 1.35 m.s^{-1} . The agent reaches this velocity within the first time-steps because a unity value has been selected for the time constant (τ). Without any other obstruction from other agents and physical obstacles the agent reaches the exit in around 225 s. At 5 and 140 ppmv (Cases A2 & A3), the agent accumulates a TL of one or higher that induces a stronger force and the agent moves faster than the desired velocity (Figure 7). Although, the

agent's health is affected by irritations etc. the evacuation completes faster (~ 160 s). Noteworthy, the agent of Case A3 cannot maintain the maximum speed, driven by the H_2S odor, and this is progressively reduced. In contrast, the agent of Case A4 is not able to reach the exit, TL approximates the highest level at around 80 s and fails completely at around 130 s. These four cases verify that the implemented software is properly following and executing the designed algorithm. Especially, they demonstrate the advantages of a dose-based evacuation algorithm against others that neglect the physical and/or psychological impact of the toxic gas on evacuation or employ a concentration-based response.

4.2 Test Cases B – Sensitivity with single agent and varying concentration

The reasoning and profound differences of coupling a dose-based response in evacuation algorithms are demonstrated profoundly in test cases B (see Figure 8). In Cases B1 and B2 the single agent is exposed at the exact same spatial varying concentration (see Figure 5) but the evacuation results are entirely different. In Case B1 (similarly to the Case A1) the agent is evacuating moving with a constant walking speed and reaches the exit at 225 s. However, the agent in Case B2 fails to exit “on time”. The agent is exposed, initially, at low concentration levels, accumulates a TL of around one, and thus moves faster than the standard velocity. In the meantime, the agent approaches the center of the toxic cloud where the higher concentrations push the TL to around two; therefore, the agent velocity decreases below the typical evacuation speed. This significant reduction on the walking speed, forces the agent to stay longer within the cloud, the TL approximates three, the walking speed decreases further, and at the end fails completely at around 280 s. Conversely, it would not have been possible to flag that evacuation is less probable without considering the impact of the toxic component.

The peak concentration of the toxic cloud in Case B2 (~ 450 ppmv) is considerably high, thus one could hypothesize that a simpler model could also flag this issue by using only the concentration levels as an index. This hypothesis is easily proven false in Case B3, where the agent evacuates through a “stronger” cloud

(peak concentration ~560 ppmv) although significantly less wide (~50 m). Initially, the agent evacuates with the typical speed and zero TL , however, the latter “jumps” instantly to around 2.5 as the agent enters the cloud. The agent’s speed reduces drastically (around 0.5 m s^{-1}) but the agent manages to exit the cloud and continue the evacuation. Regardless the severe health damage imposed on the agent, the evacuation is “successful” at around 550 s, twice as much compared with the uncoupled and unobstructed evacuation in Case B1.

4.3 Test Cases C – Realistic scenarios with multiple single agents

Figure 9 summarizes the statistical results of all the simulations executed for Case C i.e. the exit times (as box plots) and number of agents’ failing to evacuate. The blue solid box represents the reference case where either the H_2S concentration is considered zero or the evacuation calculations do not consider the toxic’s impact. At low concentration levels (4-90 ppmv) evacuation times decrease by around 30% because of the higher walking speed of some agents and regardless of the obstacles and crowd effects, attributed to the effect of the first symptom ($TL < 1$). As the impact of the second and third symptom (TL 1-3) affects the evacuation process (for concentrations 170-420 ppmv) the average exit time increases slightly but the distribution becomes wider and the maximum times increases significantly. At higher concentration levels (>420 ppmv) a number of agents fails to exit and reaches 58%. In contrast, the exit times decrease and the range of the distribution becomes narrower, due to exclusion of the failed agents from the distribution. In other words, only the agents that are close enough to exit manage to evacuate and they do so in shorter times. Indeed, only 75% of the evacuees is affected negatively by the toxic gas.

Clearly, the starting position of the agent is crucial on the outcome of the evacuation. Integration of the individual observations results to a qualitative assessment of the evacuation outcome that can be translated to evacuation zones, for this specific case, and Figure 6c illustrates these zones. According to this, 43% of the agents lie in the green area and are able to successfully evacuate the premises irrespective of the concentration levels. Evacuation of agents in the yellow region (18%) depends on the concentration levels and their overall behavior varies according to them. Finally, the agents in the red zone (39%) are the ones affected most by the toxic's levels and in many scenarios they were not able to evacuate. To mitigate the consequences, additional emergency exits, personal breathing apparatuses, or shelter-in-place could be introduced.

5 Conclusions

In this work, a fully coupled exposure-response-evacuation algorithm has been proposed with the objective to improve the evacuation modelling in the case of airborne toxics. Contrary to state-of-the-art, in this algorithm the dynamically estimated exposure of the evacuee affects, non-linearly, the evacuation speed and thus the overall process. The diversity and advantages of the methodology have been successfully demonstrated over three case studies. However, the coupling of evacuation and exposure depends on the available epidemiological and toxicological data and thus it was challenging. This is also the main hindrance validating it either directly with crowd observations as in the works of Georgoudas et al. (2011); Tsiftsis et al. (2016) or indirectly with statistical data as in Klüpfel (2007); Schadschneider et al. (2009); Ronchi et al. (2012). Despite the challenges, industrial partners can produce the required validation data by at least two plausible and feasible ways: i) analysis of video footage from past incidents, some large accidents have been reported in the introduction while many smaller ones likely have occurred as well; ii) simulated evacuation exercises using a non-toxic gas as a proxy. There is still the challenge of sharing such sensitive data with academia, but at least it is possible to validate complex algorithms like the one proposed herein. Nevertheless, the proposed algorithm could serve as an accessory for any type of evacuation model and not

specifically just for social force models. Since the algorithm ultimately aims to affect the evacuation velocity of individuals, it theoretically has the ability to manifest itself in a way that is compatible with any evacuation model that incorporates force terms, speed-density relationships (Luo et al., 2008; Zheng et al., 2016), or other models based on the Fundamental Diagram (Seyfried et al., 2007; Narang et al., 2015). This is similar to its ability to be used with any toxic gas in which the exposure-response data just needs to be replaced for the appropriate gas, similarly to what Bae and Ryou (2015) has achieved. In conclusion, it is an algorithm designed and implemented in a way to be easily adapted by safety engineers and further improved by researchers as more data become available. In any case, this work indirectly highlights the need for new toxicological data in order to improve emergency response and preparedness in the case of disasters related to airborne toxics.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

NMS: Data curation; Formal analysis; Investigation; Methodology; Visualization; Software; Writing - original draft

KEK: Conceptualization; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Visualization; Writing - review & editing

NA: Data curation; Investigation; Software; Writing - review & editing

LV: Funding acquisition; Project administration; Resources; Writing - review

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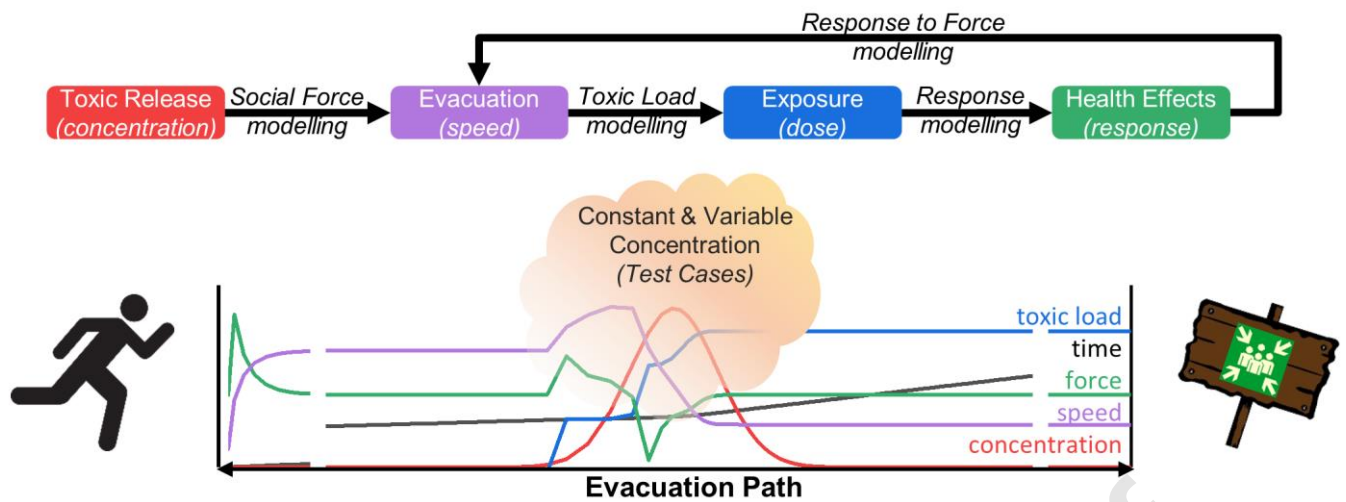


Figure 1 Illustration of the algorithm and the coupling of the three models (exposure-response-evacuation) coupling, the main variables (concentration, speed, force, toxic load), and the studied test cases (constant and variable concentration).

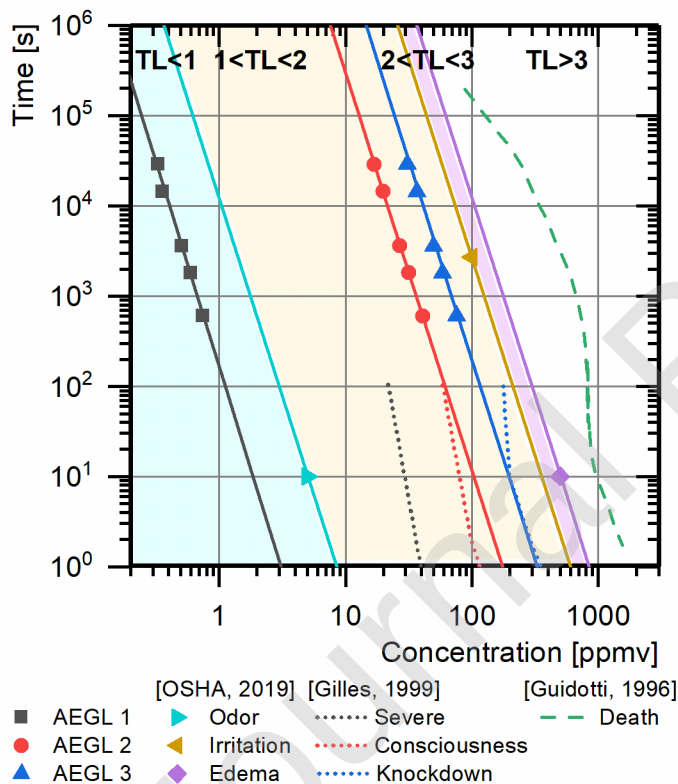


Figure 2 Literature exposure-response data for selected symptoms (Guidotti, 2010; OSHA, 2019), the AEGL values, proposed effects by Gillies et al. (1999) and the lethality curve by Guidotti (1996). Solid lines are fitted power law functions with a consistent exponent of ~ 4.4 (equation 2). TL ranges and shaded areas represent own methodology.

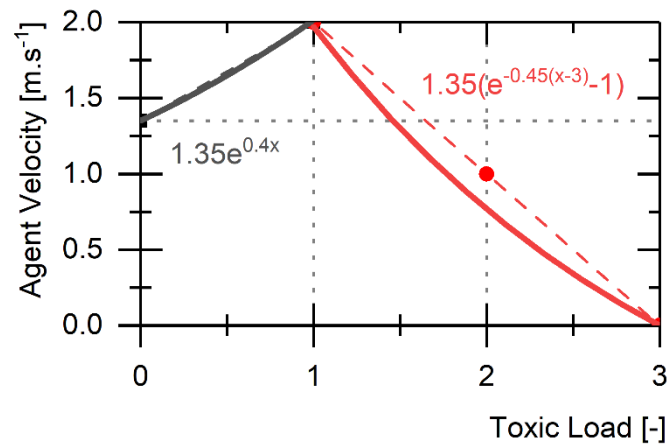


Figure 3 The fitted non-linear equation for agent's velocity (v^{tf}) as a function of the Toxic Load (TL).

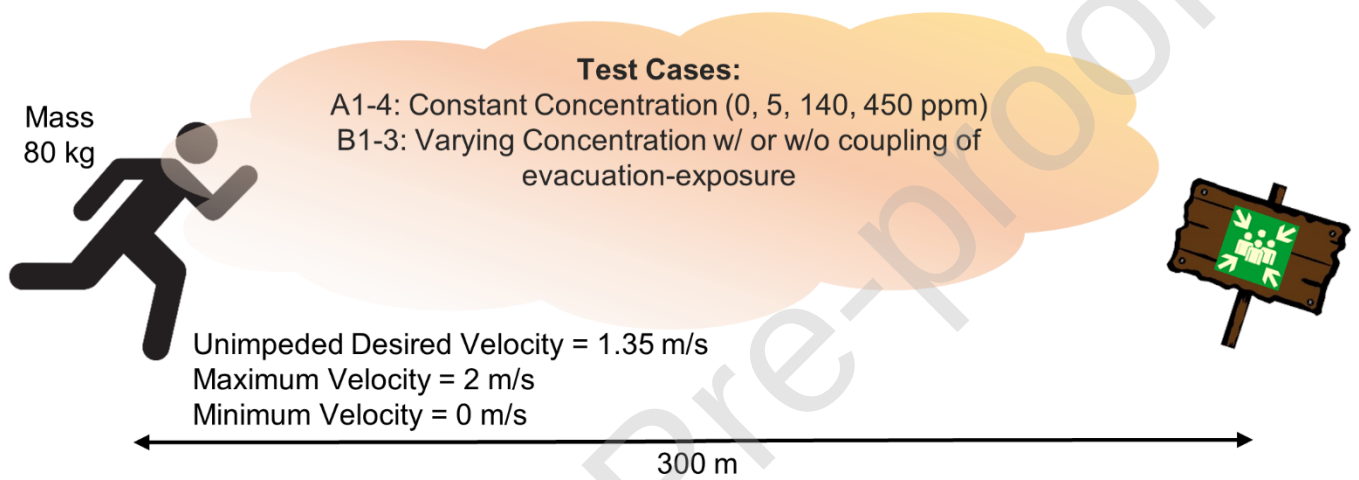


Figure 4 – Graphic of the overall conditions for the single agent test cases A and B.

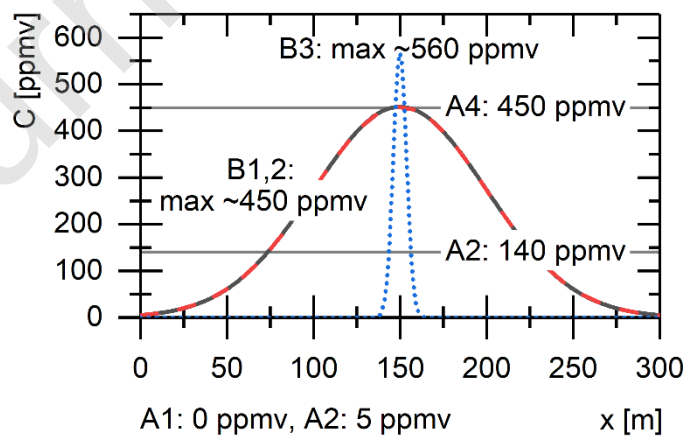


Figure 5. The concentration levels and profiles for each of the A and B Test Cases (see also Figure 4).

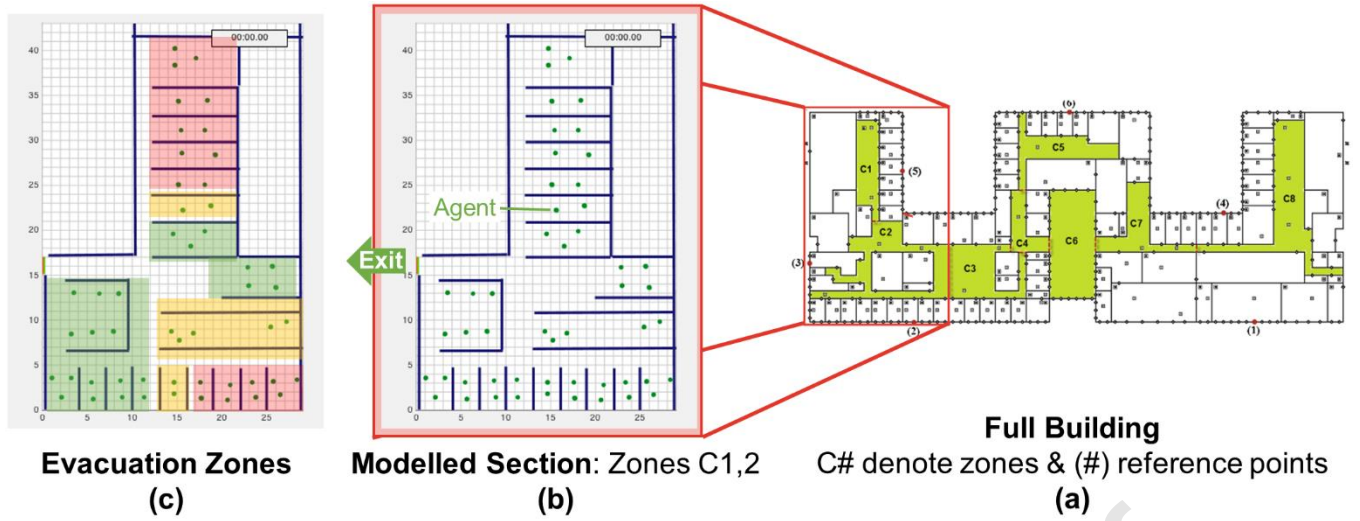


Figure 6. The evacuation scenario for Test Case C a) the full building layout, b) the modelled section of the simplified building with the 51 agents (small green circles) and a single exit, and c) the identified evacuation zones.

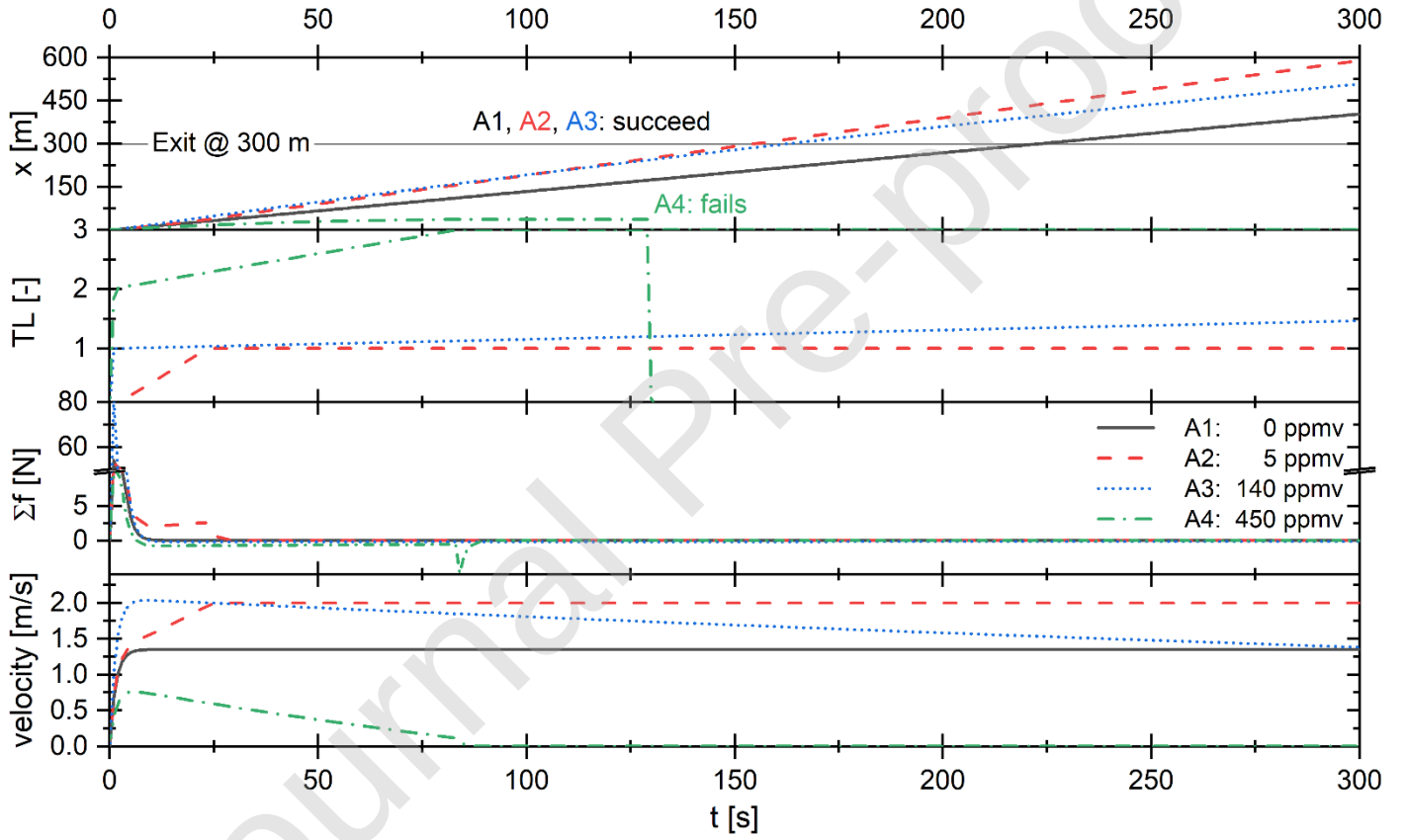


Figure 7 – Profiles of distance walked by the agent, x [m], toxic load, TL [-], total social force, f [N], and achieved speed, u [m.s⁻¹] when exposed constantly to 0, 5, 140, & 450 ppmv for the A1, 2, 3, & 4 test cases. For clarity, note the break of force axis from 10 to 60 N.

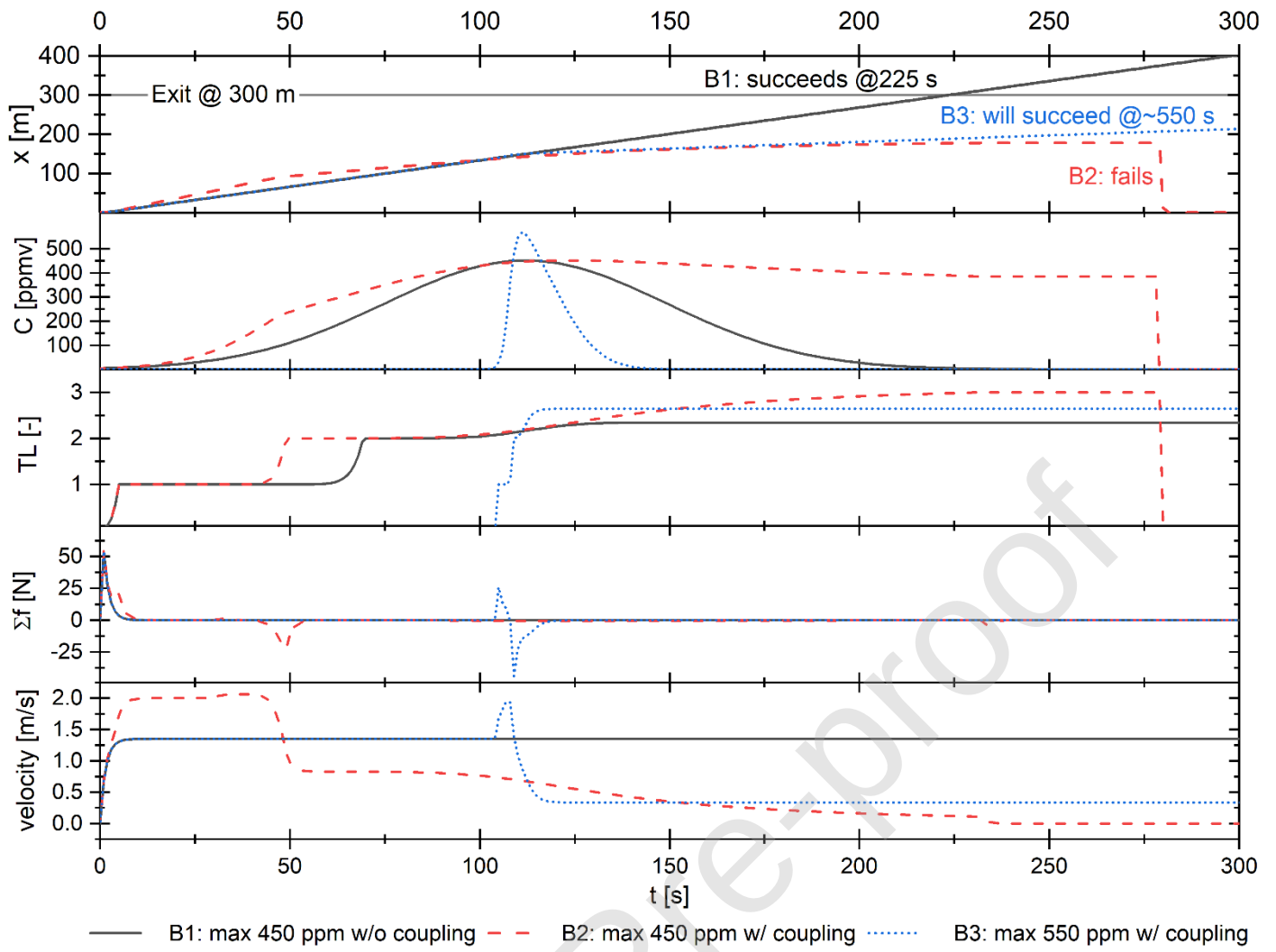


Figure 8 – Profiles of distance walked by the agent, x [m], concentration, C [ppmv] exposed to, toxic load, TL [-], total social force, f [N], and achieved speed, u [$m \cdot s^{-1}$] when exposed to varying concentration (see Figure 5) for the B1, 2, & 3 test cases.

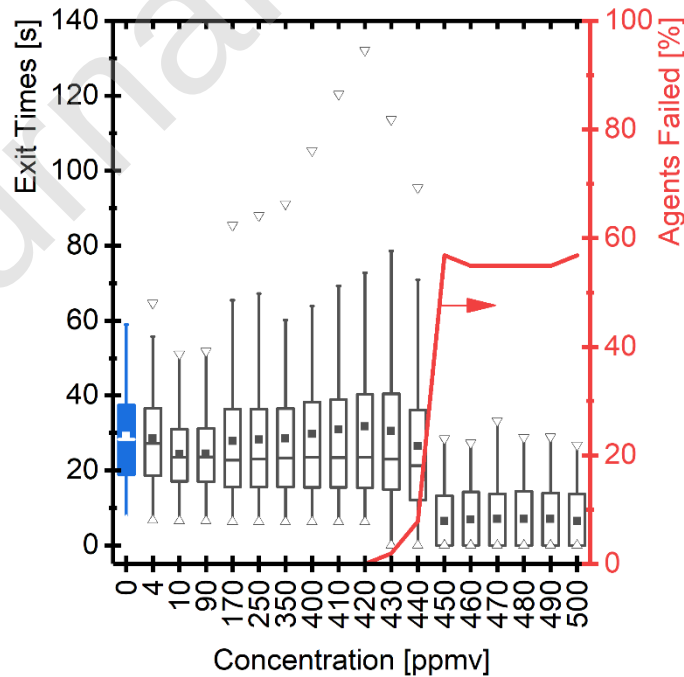


Figure 9- Typical Box plot – distribution - of agents' exit times (showing the extremes-triangles, median-horizontal line, mean-solid square, 25th and 75th percentile-box, and outliers-whiskers), and percentage of failed agents at various concentration levels for the building evacuation scenario (Test Cases C). Note the reference case C0 (blue solid box) with no exposure to the toxic gas.

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Table 1 – Selected short-term effects and “acute” concentrations from literature (Guidotti, 2010; OSHA, 2019) and in this study.

Symptoms	Concentration (ppm)	Selected dosage in this Study	Onset Toxic Load
Odor <i>threshold becomes offensive</i>	0.01-1.5 3-5	5 ppm for 10 s	1
Irritations <i>Slight conjunctivitis ("gas eye") & respiratory tract irritation (1 hour)</i> <i>Coughing, eye irritation, loss of smell (2-15 min)</i>	50-100 100	100 ppm for 30 min	2
Pulmonary edema <i>"prolonged" exposure</i>	200-300	500 ppm for 10 s	3
Knockdown <i>staggering collapse (5 min)</i> <i>immediate collapse</i>	500-700 700-1000		